(a) Negative
(b) Positive
(c) Perfect positive
(d) No correlation

4. Correlation co-efficient lies between
(a) 0 to ∞ (b) –1 to +1
(c) –1 to 0 (d) –1 to ∞

5. If $r(X,Y) = 0$ the variables X and Y are said to be
(a) Positive correlation
(b) Negative correlation
(c) No correlation
(d) Perfect positive correlation

6. The correlation coefficient from the following data N=25, ΣX=125, ΣY=100, ΣX^2=650, ΣY^2=436, ΣXY=520
(a) 0.667 (b) -0.006
(c) –0.667 (d) 0.70

7. From the following data, N=11, ΣX=117, ΣY=260, ΣX^2=1313, ΣY^2=6580, ΣXY=2827the correlation coefficient is
(a) 0.3566 (b) –0.3566
(c) 0 (d) 0.4566

8. The correlation coefficient is
(a) $r(X,Y)=\frac{\sigma_x \sigma_y}{\text{cov}(x,y)}$
(b) $r(X,Y)=\frac{\text{cov}(x,y)}{\sigma_x \sigma_y}$
(c) $r(X,Y)=\frac{\text{cov}(x,y)}{\sigma_y}$
(d) $r(X,Y)=\frac{\text{cov}(x,y)}{\sigma_x}$

9. The variable whose value is influenced (or) is to be predicted is called
(a) dependent variable
(b) independent variable
(c) regressor
(d) explanatory variable

10. The variable which influences the values or is used for prediction is called
(a) Dependent variable
(b) Independent variable
(c) Explained variable
(d) Regressed

11. The correlation coefficient
(a) $r=\pm\sqrt{b_{xy}\times b_{yx}}$
(b) $r=\frac{1}{b_{xy}\times b_{yx}}$
(c) $r=b_{xy}\times b_{yx}$
(d) $r=\pm\sqrt{\frac{1}{b_{xy}\times b_{yx}}}$

12. The regression coefficient of X on Y
(a) $b_{xy}=\frac{N\Sigma dx\,dy-(\Sigma dx)(\Sigma dy)}{N\,\Sigma dy^2-(\Sigma dy)^2}$
(b) $b_{yx}=\frac{N\Sigma\,dx\,dy-(\Sigma dx)(\Sigma dy)}{N\,\Sigma dy^2-(\Sigma dy)^2}$
(c) $b_{xy}=\frac{N\Sigma dxdy-(\Sigma dx)(\Sigma dy)}{N\,\Sigma dx^2-(\Sigma dx)^2}$
(d) $b_{y}=\frac{N\Sigma xy-(\Sigma x)(\Sigma y)}{\sqrt{N\Sigma x^2-(\Sigma x)^2}\times\sqrt{N\Sigma y^2-(\Sigma y)^2}}$

13. The regression coefficient of Y on X
(a) $b_{xy}=\frac{N\Sigma dx\,dy-(\Sigma dx)(\Sigma dy)}{N\,\Sigma dy^2-(\Sigma dy)^2}$
(b) $b_{yx}=\frac{N\Sigma dxdy-(\Sigma dx)(\Sigma dy)}{N\Sigma dy^2-(\Sigma dy)^2}$
(c) $b_{yx}=\frac{N\Sigma dx\,dy-(\Sigma dx)(\Sigma dy)}{N\Sigma dx^2-(\Sigma dx)^2}$
(d) $b_{xy}=\frac{N\Sigma xy-(\Sigma x)(\Sigma y)}{\sqrt{N\Sigma x^2-(\Sigma x)^2}\times\sqrt{N\Sigma y^2-(\Sigma y)^2}}$

14. When one regression coefficient is negative, the other would be
(a) Negative (b) Positive
(c) Zero (d) None of them

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